CONSTRUCTION PLANNING AGENT

A Project-II Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

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Faculty of Engineering

MEDICAPS UNIVERSITY, INDORE- 453331

JAN-JUNE 2026

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Report Approval

The project work **"Construction planning Agent"** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner

Name:

Designation

Affiliation

Declaration

I/We hereby declare that the project entitled **"Construction Planning Agent"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in 'B.Tech. Computer Science' completed under the supervision of **Dr. Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena,** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

Certificate

I/We, **Dr. Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena** certify that the project entitled **"Construction Planning Agent"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Ansh Kumbhare, Anshul Sen, Anshul Yadav, Armi Bhawsar, Atharv Dubey** is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Prof. Akshay Saxena	<Name of External Guide (If any)>
Computer Science and Engineering	Computer Science and Engineering
Medicaps University, Indore	Datagami Technology Services pvt/ltd

Dr. Kailash Chandra Bandhu

Head of the Department

Computer Science & Engineering

Medicaps University, Indore

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I express my heartfelt gratitude to my **External Guide, Mr. Suraj Naik, Mr. Ashwin Krishnan and Mr. Vaibhav, Project Lead, Datagami** as well as to my Internal Guide**, Prof. Akshay Saxena, Prof. Ajeet Singh Rajput, Dr. Hemlata Patel, Professor, Department of Computer Science Engineering, Medicaps University, Indore** without whose continuous help and support, this project would ever have reached to the completion.

It is their help and support, due to which we became able to complete the design and technical report.

Without their support this report would not have been possible.

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Abstract

The construction industry involves complex planning processes that require efficient management of time, resources, cost, and risks. Traditional planning methods are often manual, time-consuming, and prone to inefficiencies. With the advancement of Artificial Intelligence, there is a growing need for intelligent systems that can automate and optimize construction planning.

This project presents the development of an **Integrated Construction Planner Agent**, an AI-powered system designed to assist in construction project planning and decision-making. The system utilizes a **multi-agent architecture**, where a central intelligent planner processes user inputs and decomposes them into specific tasks such as scheduling, resource allocation, cost estimation, and risk analysis. These tasks are handled by specialized agents, and the results are aggregated to generate a comprehensive construction plan.

The application is built using modern technologies including **Large Language Models (LLMs)** for understanding user queries and generating insights, along with a backend system for orchestration and data handling. A user-friendly interface enables users to input project requirements and receive structured outputs in real time.

The proposed system improves planning efficiency, reduces manual effort, and enhances decision-making by providing integrated and intelligent solutions. It demonstrates the potential of AI-driven approaches in transforming traditional construction management practices into more automated, scalable, and data-driven systems.

This work demonstrates the practical application of AI-driven multi-agent systems in construction management and highlights their potential to transform traditional planning processes into intelligent, automated, and data-driven solutions.

Keywords:

Artificial Intelligence, Construction Planning, Multi-Agent Systems, Large Language Models, Project Scheduling, Resource Management, Cost Estimation, Risk Analysis

Table of Contents

		Page No.
	Report Approval	ii
	Declaration	iii
	Certificate	iv
	Acknowledgement	v
	Abstract	vii
	Table of Contents	viii
	List of figures	ix
	List of tables	x
	Abbreviations	xi
	Notations & Symbols	xii
Chapter 1	Introduction (Whichever is applicable)	
	1.1 Introduction	1
	1.2 Literature Review	1
	1.3 Objectives	2
	1.4 Significance	2
	1.5 Research Design	2
	1.6 Source of Data	3
	1.7 Chapter Scheme	4
Chapter 2	Present Investigation	
	2.1 Experimental Set-up	6
	2.2 Procedures Adopted	7
Chapter 3	System Design	
	3.1 Application Design	8
	3.2 Process Flow	9
	3.3 Information Flow	10
	3.4 Components Design	11
Chapter 4	Implementation	
	4.1 Development Environment	12
	4.2 Technology Stack	12
	4.3 Module-wise Implementation	12
	4.4 API Integration	13
Chapter 5	Devops and Deployment	
	5.1 Dockerization	14
	5.2 CI/CD Pipeline	14
	5.3 AWS EC2 Deployment	15
Chapter 6	Results and Discussions	

	6.1 System Output	16
	6.2 Challenges Faced	18
Chapter 7	Summary and Conclusions	
	7.1 Summary	19
	7.2 Conclusions	19
Chapter 8	Future scope	
	Appendix	20
	Bibliography	21

List of Figures

Figure 3.1: Application Design of Construction Planning Assistant Agent
Figure 3.2: Process Flow Diagram of the System
Figure 3.3: Information Flow Between Modules
Figure 3.4: Component-wise System Design Architecture

List of Tables

Table 1.2: Sources of Data Used in the System

Table 2.1: Experimental Setup Configuration

Abbreviations

The following abbreviations are used in the Construction Planning Assistant Agent project report:

- AI – Artificial Intelligence
- LLM – Large Language Model
- API – Application Programming Interface
- UI – User Interface
- EC2 – Elastic Compute Cloud (AWS service)
- AWS – Amazon Web Services
- CI/CD – Continuous Integration and Continuous Deployment
- JSON – JavaScript Object Notation
- Docker – Containerization platform for application deployment
- Git – Version control system
- GitHub – Cloud-based Git repository hosting service
- ML – Machine Learning

Notations & Symbols

The following notations and symbols are used in the Construction Planning Assistant Agent project report for better understanding of system components and processes:

- AI / LLM – Artificial Intelligence / Large Language Model used for natural language understanding and planning.
- UI – User Interface (Streamlit-based frontend).
- API – Application Programming Interface used for communication between modules.
- EC2 – Amazon Elastic Compute Cloud used for deployment.
- CI/CD – Continuous Integration and Continuous Deployment pipeline for automation.
- Docker – Containerization platform used to package the application.
- Agent – Independent functional module performing specific tasks (Scheduling, Cost, Resource, Risk).
- JSON – Data format used for structured communication between modules.

Chapter-1

1.1 Introduction

The construction industry involves complex planning processes that require coordination among multiple tasks, resources, and stakeholders. Effective planning is essential to ensure projects are completed within time, budget, and quality constraints. However, traditional construction planning methods are often manual, time-consuming, and prone to human errors, leading to delays, cost overruns, and inefficiencies in project execution.

With the advancement of Artificial Intelligence (AI) and Large Language Models (LLMs), there is significant potential to automate and optimize construction planning. AI-driven systems can analyze complex requirements, process large volumes of data, and generate structured plans, reducing dependency on manual effort and improving decision-making.

This project, **Construction Planning Assistant Agent,** aims to develop an intelligent system that generates structured construction plans from user inputs. The system uses an agent-based approach to analyze requirements, break down complex tasks into smaller components, and produce detailed execution strategies. This approach ensures systematic, modular, and adaptable planning.

Furthermore, modern AI models have demonstrated the ability to understand complex instructions and convert them into practical, structured solutions. By integrating planning components into a unified system, the proposed solution enhances coordination, improves accuracy, and reduces inconsistencies.

Overall, the system focuses on automating construction planning, improving efficiency, and minimizing human effort, while leveraging modern technologies to ensure scalability and real-world applicability.

1.2 Literature Review

The field of construction planning has evolved with the integration of digital tools and Artificial Intelligence. Traditional methods like Gantt charts help in scheduling but lack intelligent decision-making. Modern AI-based systems use algorithms and agent-based approaches to break down complex tasks, generate workflows, and improve planning efficiency.

Research in multi-agent systems shows that tasks can be divided among intelligent agents to handle complex projects effectively. Additionally, Large Language Models (LLMs) can understand user inputs and generate structured, step-by-step plans, making them useful in construction and engineering domains.

AI tools are also being used to automate documentation and requirement analysis. However, challenges such as handling real-world constraints, maintaining accuracy, and adapting to dynamic changes still remain. This project addresses these issues by integrating AI agents into construction planning for a smarter and more efficient solution.

1.3 Objectives

The main objectives of the Construction Planning Agent project are:

- To develop an AI-based system for automating construction planning
- To convert user requirements into structured task-wise plans
- To reduce manual effort and human errors in planning
- To improve efficiency and accuracy of construction workflows
- To implement an agent-based approach for task decomposition and execution
- To generate scalable and adaptable planning solutions for different types of construction projects

1.4 Significance of the Study

This project is significant as it addresses major challenges in traditional construction planning methods.

- Helps in reducing time and cost overruns
- Improves decision-making using AI-based insights
- Provides automation in planning processes
- Enhances productivity and workflow efficiency
- Supports modern digital transformation in construction industry
- Demonstrates the application of AI and multi-agent systems in real-world problems

Overall, the system contributes to the development of smart construction management solutions.

1.5 Research Design

- Based on an **applied research and system development approach**
- Focuses on designing and implementing an AI-based construction planning system
- **Problem Identification:** Analyze limitations of traditional construction planning methods
- **Requirement Analysis:** Identify user needs, inputs, and system functionalities
- **System Design:** Plan architecture including frontend, backend, and AI agent modules
- **Development:** Build the system using suitable programming tools and frameworks
- **AI Integration:** Implement agent-based logic for task decomposition and planning
- **Testing:** Evaluate system performance, accuracy, and usability
- **Deployment:** Use Docker and CI/CD pipelines for scalable and automated deployment
- **Evaluation:** Analyze results to validate effectiveness of the proposed system

1.6 Source of Data

The data used in this project is derived from multiple sources to support the development and testing of the system. These include:

- **User Inputs:** Project requirements, task descriptions, timelines, and resource details provided by users
- **Simulated Data:** Artificial datasets created to test different construction planning scenarios
- **System-Generated Data:** Outputs produced by the AI agent during planning and execution

The combination of these data sources ensures that the system is flexible and capable of handling various types of construction planning problems. Proper preprocessing and structuring of data are performed to maintain consistency and reliability.

S. No	Data Source	Description
1	User Input	Natural language construction requirements provided by user
2	LLM Models	AI-generated reasoning and planning outputs
3	Predefined Rules	Cost estimation and resource logic rules
4	System Logs	Intermediate outputs from agents for processing

Table 1.2: Sources of Data

1.7 Chapter Scheme

The project report is organized into the following chapters to provide a structured and systematic presentation of the work:

Chapter 1: Introduction

This chapter presents an overview of the project, including the background, literature review, objectives, significance, research design, source of data, and chapter scheme. It provides the foundation for understanding the need for the Construction Planning Assistant Agent and defines the scope of the system.

Chapter 2: Present Investigation

This chapter describes the experimental setup and the procedures adopted for developing and testing the Construction Planning Assistant Agent. It explains how the system was implemented and evaluated in a structured environment.

Chapter 3: System Design

This chapter explains the overall system design and architecture, including application design, process flow, information flow, and component design. It describes how different modules such as frontend, backend, and AI agents interact with each other.

Chapter 4: Implementation

This chapter focuses on the development details of the system, including the development environment, technology stack, module-wise implementation, and API integration used in the Construction Planning Assistant Agent.

Chapter 5: DevOps and Deployment

This chapter explains the DevOps practices used in the project, including Dockerization, CI/CD pipeline, and deployment on AWS EC2. It describes how the system is made scalable and production-ready.

Chapter 6: Results and Discussions

This chapter presents the system output and discusses the performance of the Construction Planning Assistant Agent. It also highlights the challenges faced during development and how they were resolved.

Chapter 7: Summary and Conclusions

This chapter provides a summary of the complete project and presents the final conclusions drawn from the implementation and results of the system.

Chapter 8: Future Scope

This chapter describes possible future enhancements of the Construction Planning Assistant Agent, including improvements in AI models, scalability, real-time integration, and advanced features.

Chapter-2 Present Investigation

2.1 Experimental Set-up

The experimental setup for the proposed system is designed to develop and evaluate an AI-based construction planning agent. The system is implemented using a combination of frontend and backend technologies along with AI models for intelligent task planning.The backend is responsible for processing user inputs, managing logic, and interacting with the AI agent. The frontend provides a user-friendly interface where users can input project requirements and view generated plans. The system is containerized using Docker to ensure portability and consistency across different environments. CI/CD pipelines are used to automate building, testing, and deployment processes. The setup also includes testing environments where different construction scenarios are simulated to evaluate system performance.

Component	Details
Operating System	Windows / Linux
Programming Language	Python 3.x
Frameworks	Streamlit, FastAPI/Flask
AI Model	Large Language Models (LLM)
Version Control	Git & GitHub
Deployment	AWS EC2

Table 2.1: Experimental Setup

2.2 Procedures Adopted

The following procedures are adopted for the development and evaluation of the system:

- Collecting and analyzing user requirements for construction planning
- Designing the system architecture including frontend, backend, and AI modules
- Developing the application using appropriate programming tools and frameworks
- Integrating the AI agent to process inputs and generate structured plans
- Testing the system with different construction scenarios and inputs
- Evaluating the accuracy, efficiency, and usability of the generated plans
- Deploying the system using Docker and CI/CD pipelines
- Analyzing results and refining the system based on performance feedback

Chapter 3 System Design

3.1 Application Design

The application follows a multi-layered agent-based architecture where user inputs are processed through an AI planner, decomposed into tasks, and executed via specialized modules.

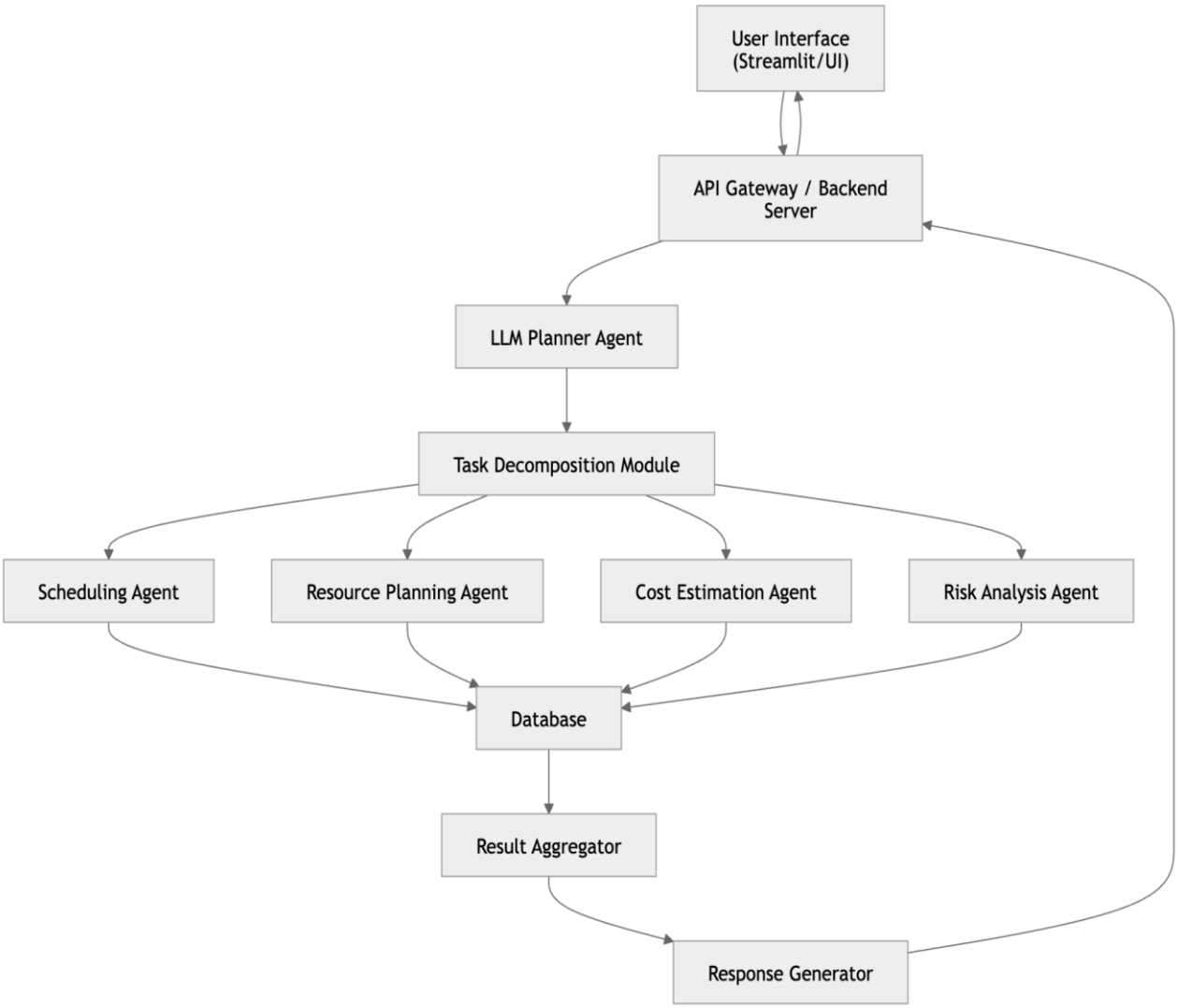


Fig 3.1 Application Design

3.2 Process Flow

This defines the **step-by-step execution pipeline** from user input to final output.

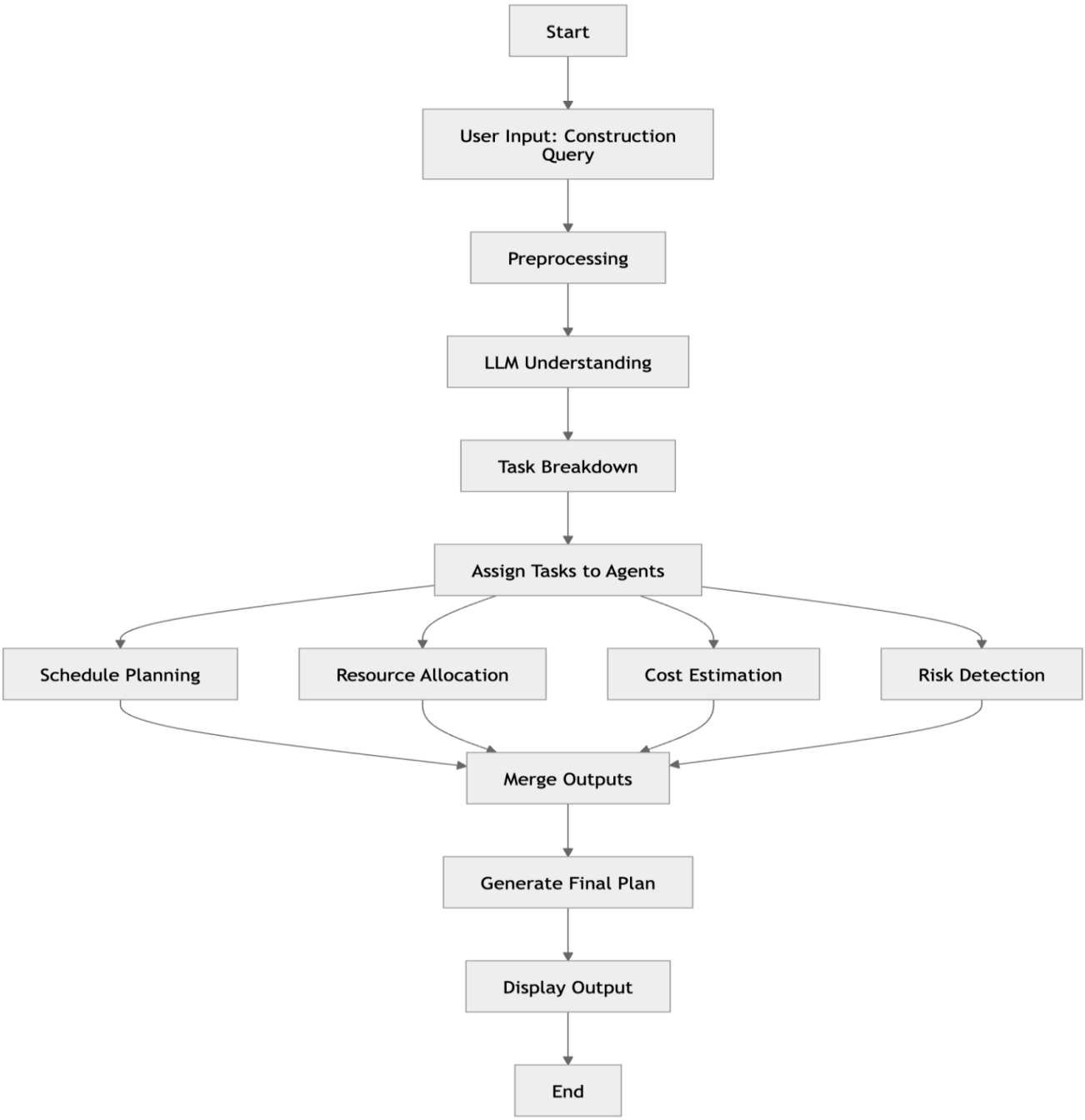


Fig 3.2 Process Flow

3.3 Information Flow

This shows how data moves across components including inputs, intermediate outputs, and final results.

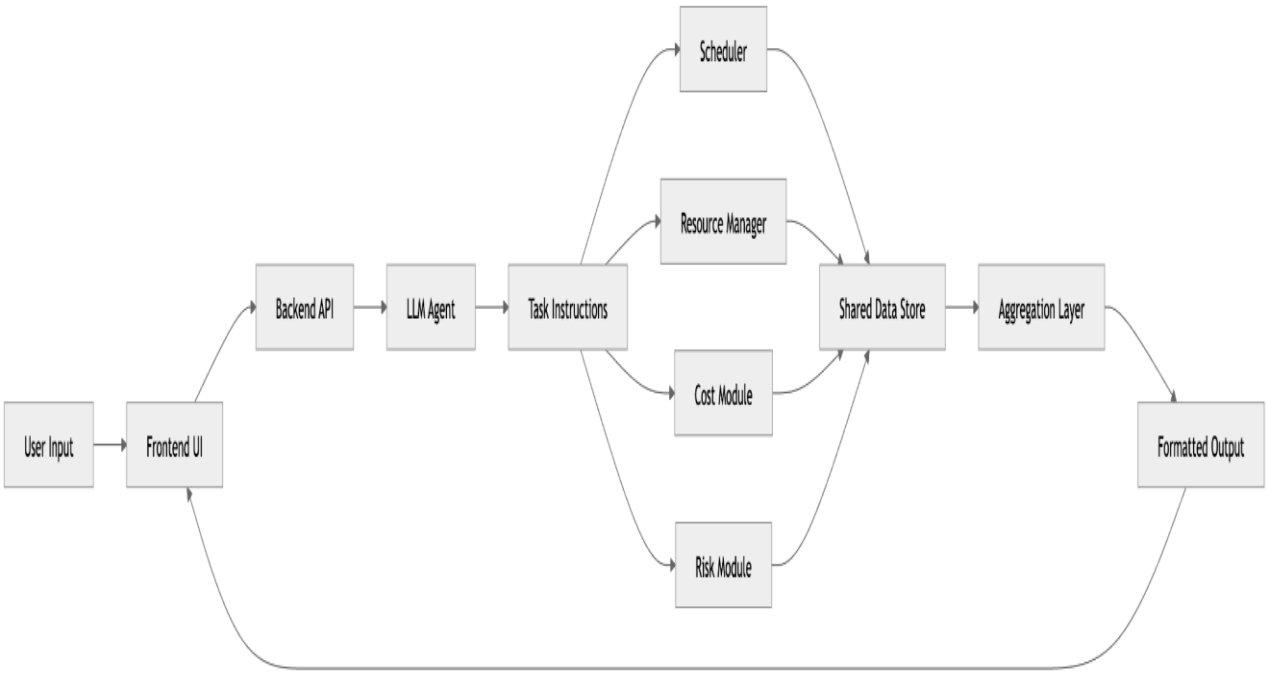


Fig 3.3 Information Flow

3.4 Components Design

Each component is modular and loosely coupled to ensure scalability and maintainability.

Key Components

- Frontend (Streamlit UI) – User interaction
- Backend API (FastAPI/Flask) – Handles requests
- LLM Planner Agent – Core intelligence (task understanding)
- Task Decomposer – Breaks query into subtasks
- Domain Agents:
    - Scheduling Agent
    - Resource Agent
    - Cost Agent
    - Risk Agent
- Database – Stores project data
- Aggregator – Combines outputs
- Response Generator – Final output formatting

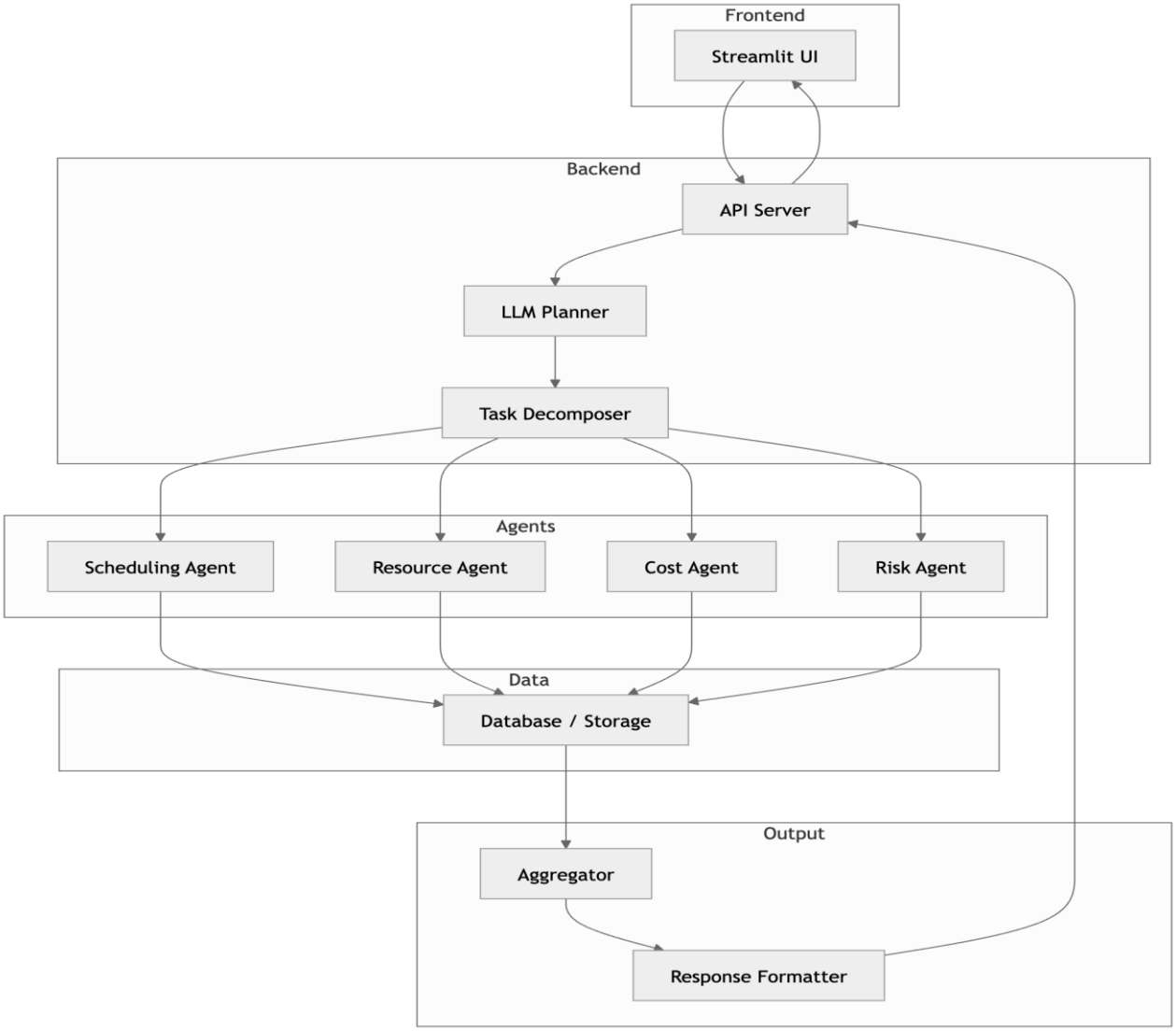


Fig 3.4 Component Design

Chapter 4 Implementation

4.1 Development Environment

The development of the Integrated Construction Planner Agent was carried out in a structured and efficient environment to ensure smooth integration of all components.

- Operating System: Windows / Linux
- Development Tools: Visual Studio Code (VS Code)
- Programming Language: Python (due to strong AI/ML ecosystem)
- Version Control: Git and GitHub (for code management and collaboration)
- Environment Management: Virtual Environment (venv) for dependency isolation
- Runtime: Python 3.x

The system was tested locally and can be extended to cloud or container-based environments for deployment.

4.2 Technology Stack

The system is developed using modern tools and frameworks:

- Programming Language: Python
- Frontend: Streamlit (for interactive UI)
- Backend: FastAPI / Flask (API handling and orchestration)
- AI Engine: Large Language Models (LLMs)
- Agent Framework: Multi-agent orchestration (CrewAI / custom agents)
- Database: JSON / lightweight storage (or optional DB)
- Deployment Tools: Docker, Kubernetes (optional based on setup)

4.3 Module-wise Implementation

The Integrated Construction Planner Agent is implemented in a modular way to ensure clarity, scalability, and easy maintenance. Each module is responsible for a specific function within the system.

1. User Interface Module

This module is developed using Streamlit and provides an interactive interface for users. It accepts natural language input related to construction planning and displays the generated outputs such as schedules, cost estimates, and risk analysis in a structured format.

2. Input Processing Module

This module preprocesses the user input by cleaning and structuring the text. It ensures that the input is properly formatted before being sent to the LLM-based planner for further processing.

3. LLM Planner Module

This is the core intelligence module of the system. It interprets the user query using a Large Language Model and converts it into structured planning tasks. It identifies key requirements such as timeline, resources, cost, and risks.

4. Task Decomposition Module

This module breaks the main planning task into smaller sub-tasks. These sub-tasks are then assigned to specialized agents for execution.

5. Agent Modules

The system includes multiple specialized agents:

- **Scheduling Agent** for project timelines
- **Resource Agent** for material and labor allocation
- **Cost Agent** for budget estimation
- **Risk Agent** for identifying potential risks

Each agent processes its assigned task independently.

6. Aggregation Module

This module collects outputs from all agents and combines them into a unified and structured construction plan.

7. Output Generation Module

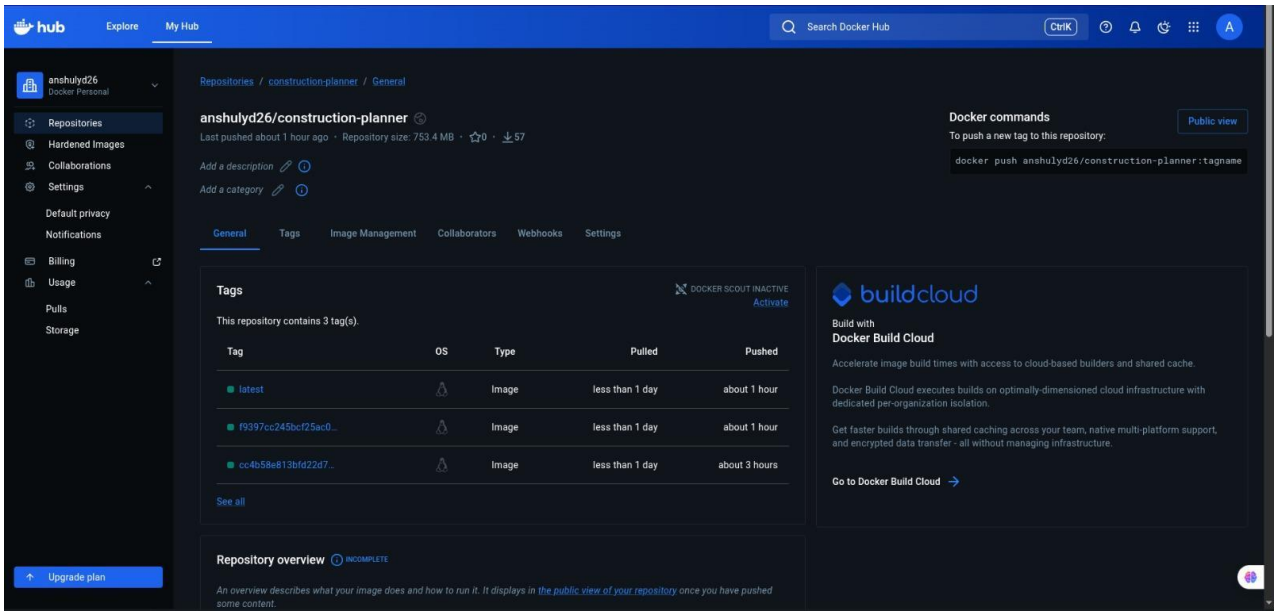
The final module formats the aggregated results and presents them in a user-friendly manner through the interface.

4.4 API Integration

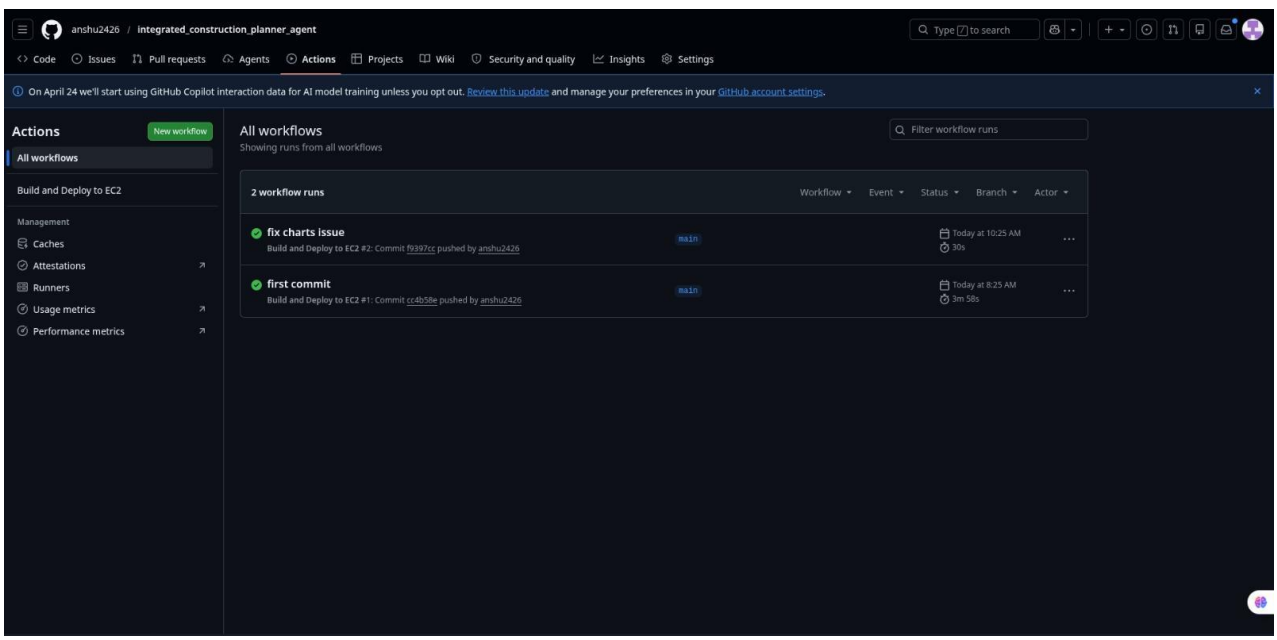
The Integrated Construction Planner Agent uses API integration to enable smooth communication between the frontend interface, backend services, and AI-based modules. The system follows a REST-based architecture where the user input from the Streamlit interface is sent to the backend through API requests. The backend processes these requests and forwards the data to the LLM Planner Module, which generates structured tasks based on the input. These tasks are then distributed to different agent modules such as scheduling, resource planning, cost estimation, and risk analysis. Each module processes its assigned task and returns the results to the backend through internal API calls. The backend then aggregates all responses and sends the final output back to the frontend for display. In addition, LLM APIs are used to handle natural language understanding and generate intelligent responses. This API-based communication ensures modularity, scalability, and efficient interaction between all components of the system.

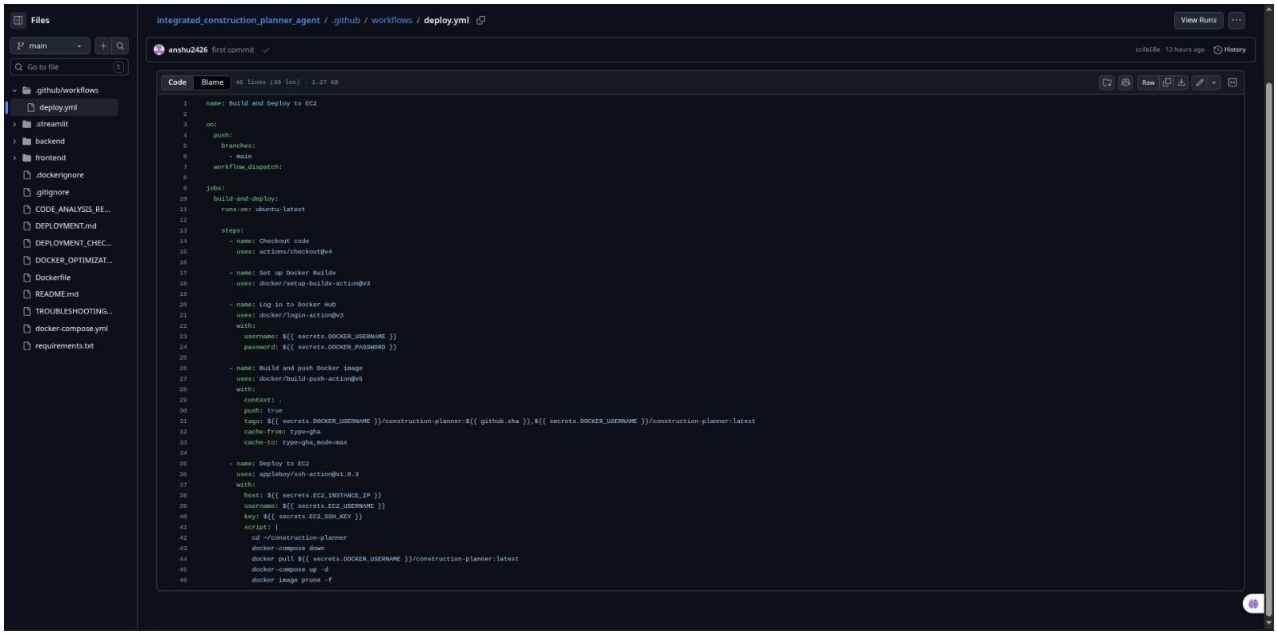
Chapter 5 Devops and Deployment

5.1 Dockerization

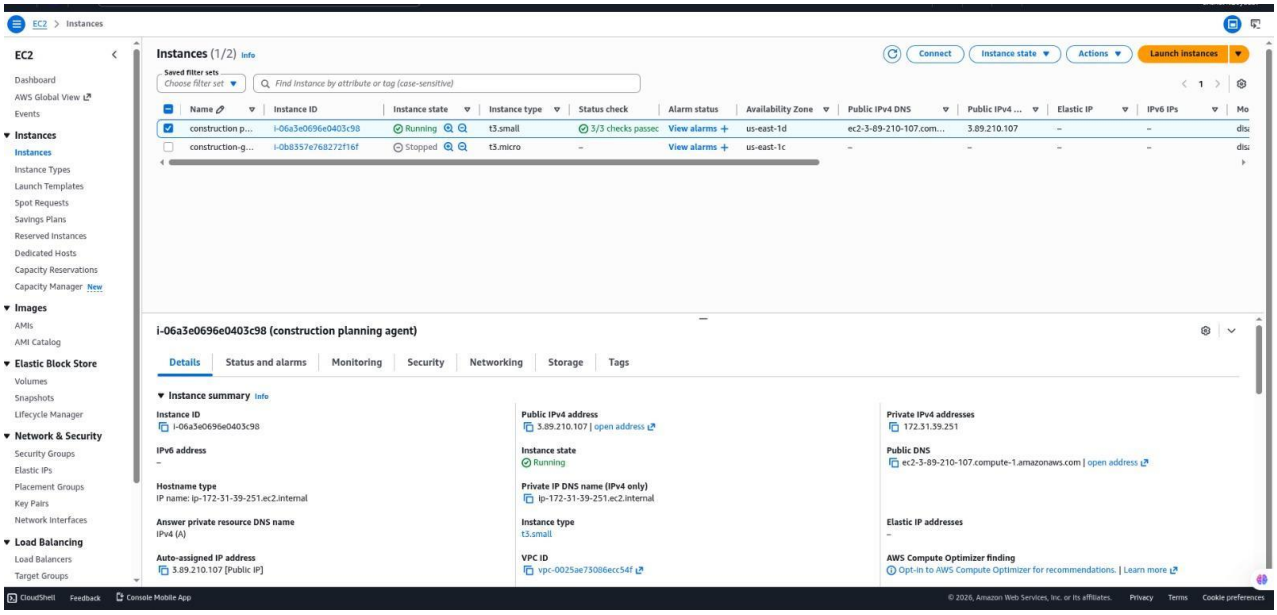


5.2 CI/CD Pipeline



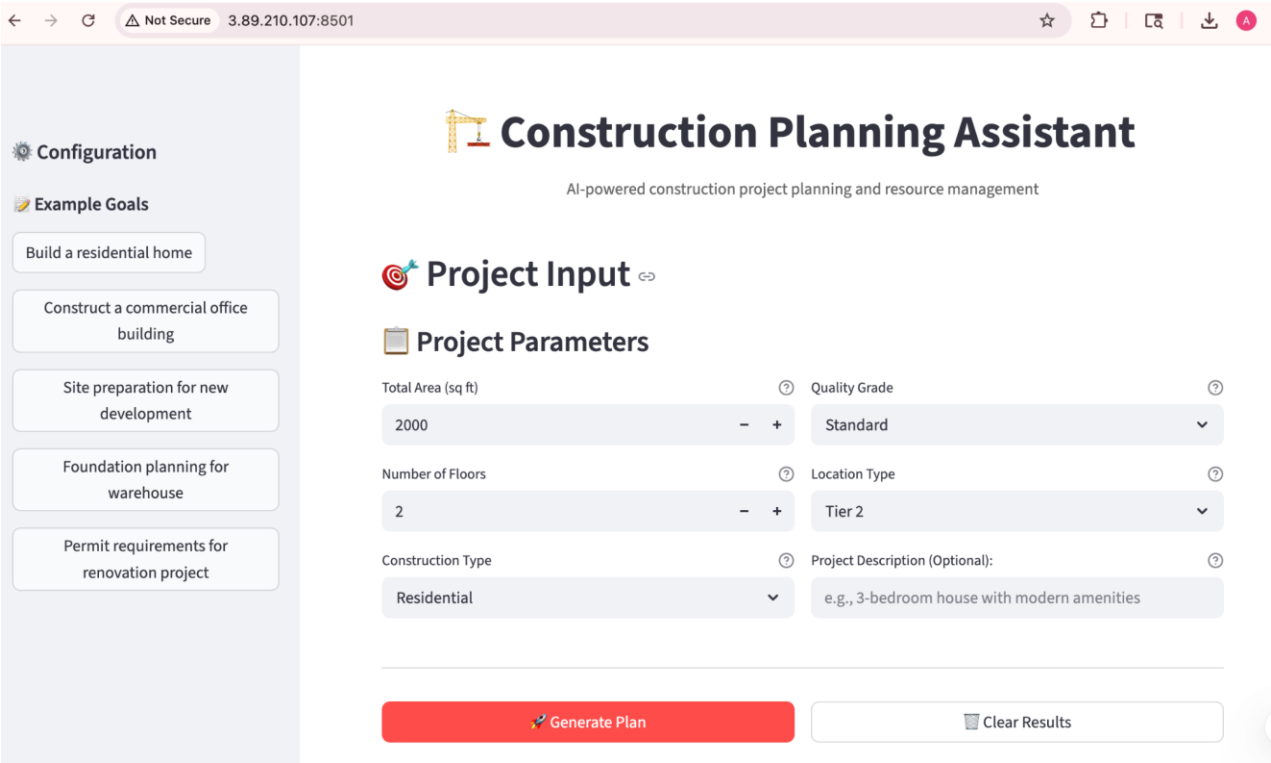


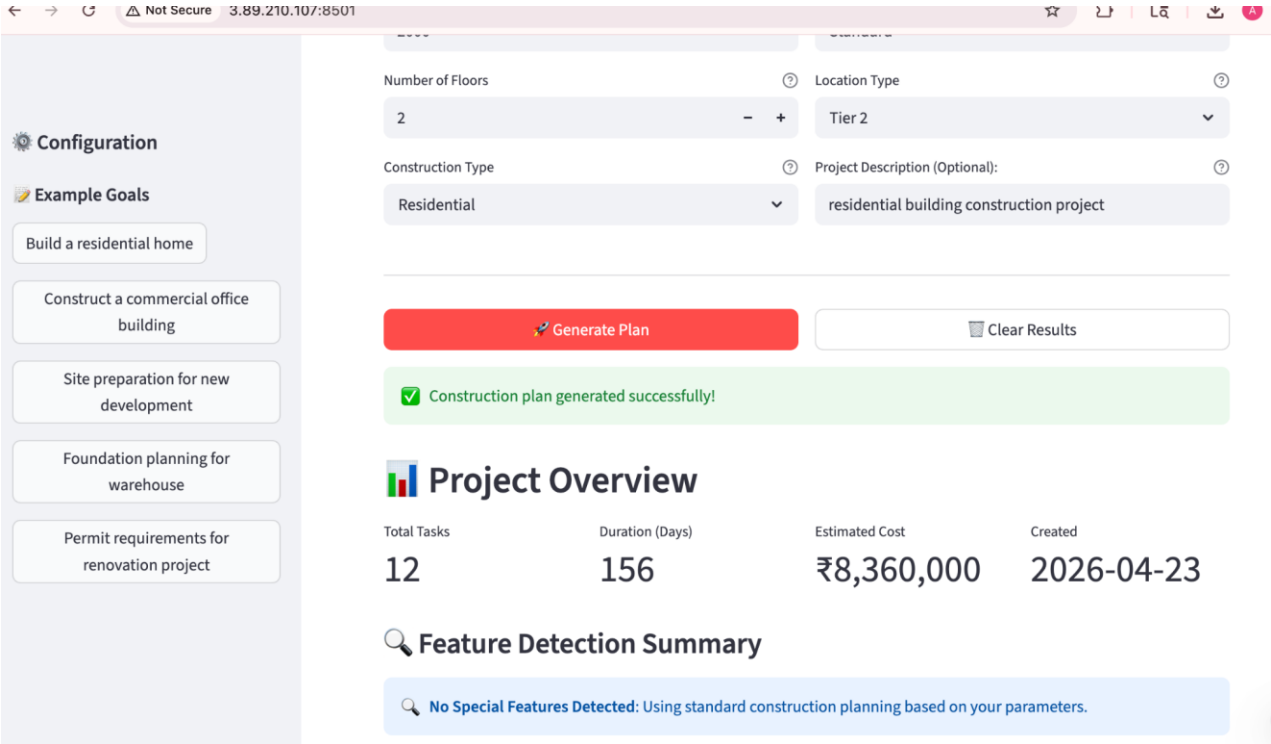
5.3 AWS EC2 Deployment

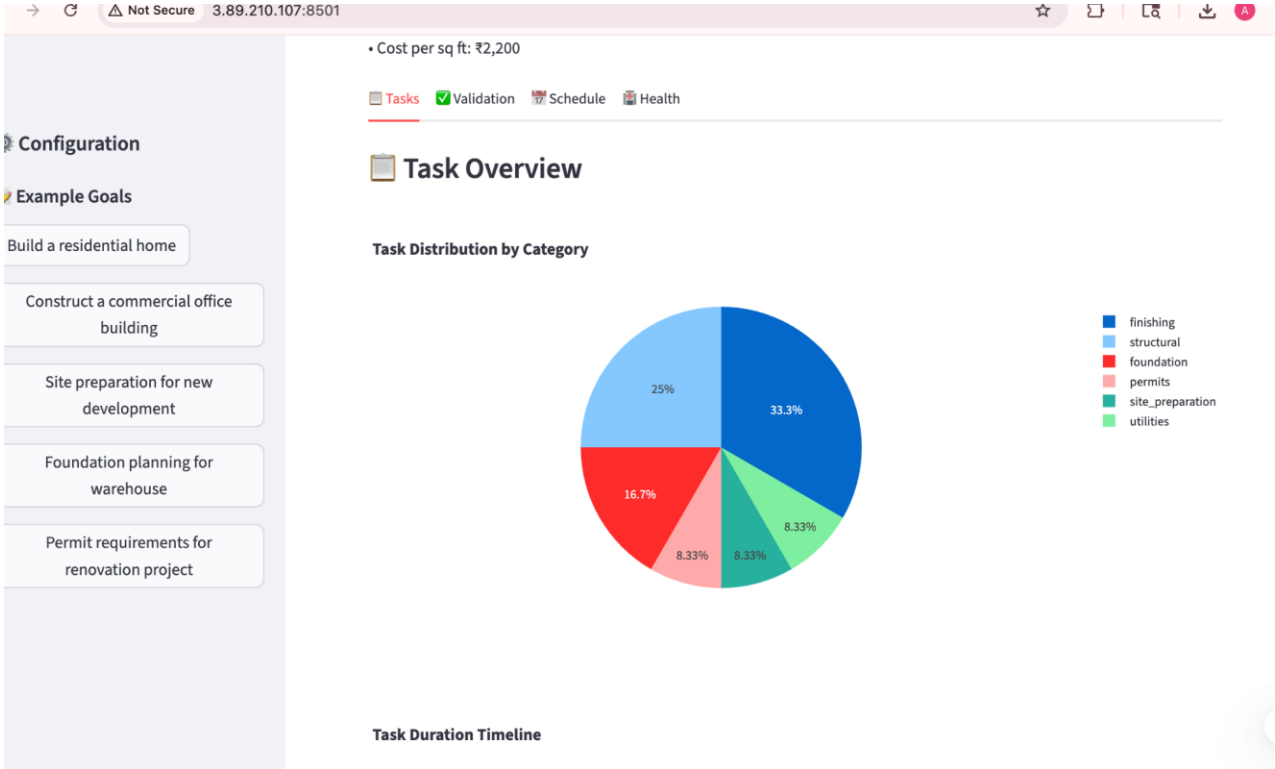


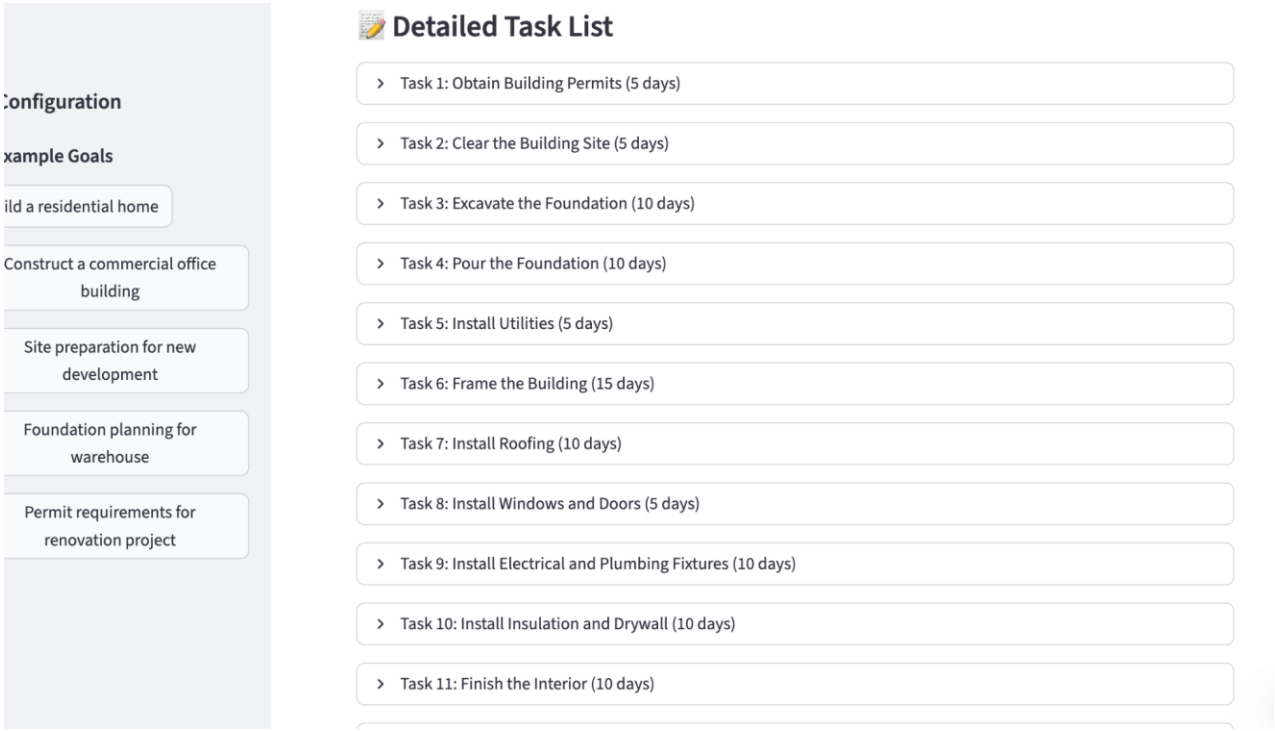
Chapter 6 Results and Discussions

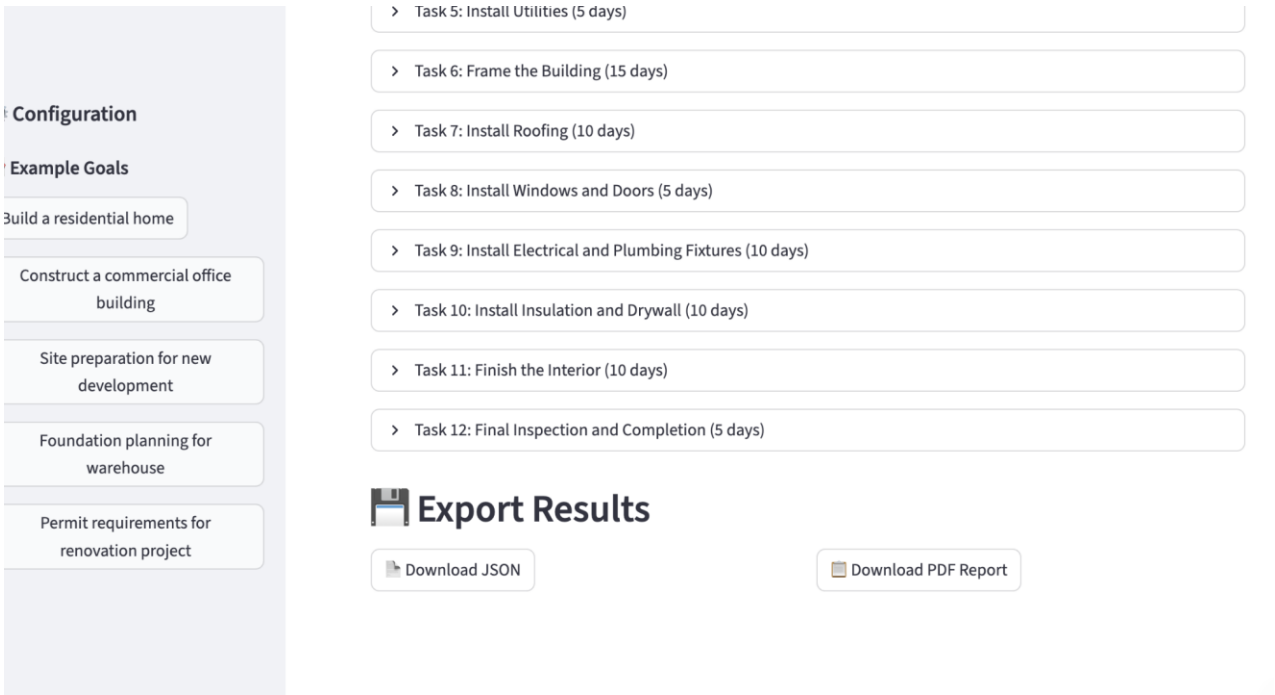
6.1 System Output






Not Secure 3.89.210.107:8501
• Cost per sq ft: ₹2,200
Tasks
Validation
Schedule
Health
Configuration
Example Goals
Build a residential home
Construct a commercial office building
Site preparation for new development
Foundation planning for warehouse
Permit requirements for renovation project
Task Overview
Task Distribution by Category
25%
33.3%
16.7%
8.33%
8.33%
8.33%
finishing
structural
foundation
permits
site_preparation
utilities
Task Duration Timeline


Configuration
Example Goals
Build a residential home
Construct a commercial office building
Site preparation for new development
Foundation planning for warehouse
Permit requirements for renovation project
Detailed Task List
Task 1: Obtain Building Permits (5 days)
Task 2: Clear the Building Site (5 days)
Task 3: Excavate the Foundation (10 days)
Task 4: Pour the Foundation (10 days)
Task 5: Install Utilities (5 days)
Task 6: Frame the Building (15 days)
Task 7: Install Roofing (10 days)
Task 8: Install Windows and Doors (5 days)
Task 9: Install Electrical and Plumbing Fixtures (10 days)
Task 10: Install Insulation and Drywall (10 days)
Task 11: Finish the Interior (10 days)



6.2 Challenges Faced

During the development of the Construction Planning Assistant Agent, several technical and integration challenges were encountered. One of the main challenges was managing the coordination between multiple AI agents, as each agent (scheduling, resource, cost, and risk) needed to work independently while still producing consistent and integrated outputs. Ensuring smooth communication between the frontend, backend, and LLM modules was another difficulty due to complex API interactions.

Another challenge was optimizing response time, as LLM-based processing and multiple agent calls increased latency. Overall, these challenges were resolved through modular design, proper API structuring, and iterative testing, resulting in a stable and efficient system.

Chapter 7 Summary and Conclusions

7.1 Summary

The **Construction Planning Assistant Agent** is an AI-based system designed to automate and improve construction project planning. The system integrates Large Language Models with a multi-agent architecture to handle key planning tasks such as scheduling, resource allocation, cost estimation, and risk analysis. The application provides a user-friendly interface through which users can input project requirements in natural language and receive structured planning outputs. The system is implemented using modular components, ensuring scalability, flexibility, and easy maintenance. It is also deployed on AWS EC2 with Docker and CI/CD support for efficient and reliable operation.

It integrates a multi-agent architecture with Large Language Models to interpret user inputs and generate structured outputs for scheduling, resource allocation, cost estimation, and risk analysis. The system provides a simple and interactive interface that allows users to describe project requirements in natural language. The backend processes these inputs and coordinates between different agents to produce a complete construction plan. The application is designed using a modular approach and deployed on AWS EC2, ensuring scalability, accessibility, and reliable performance.

7.2 Conclusion

The project successfully demonstrates the application of artificial intelligence and multi-agent systems in the field of construction planning. The **Construction Planning Assistant Agent** reduces manual effort, improves accuracy, and enhances decision-making by automating complex planning processes. The integration of LLMs and specialized agents enables intelligent decomposition and execution of tasks, resulting in efficient project planning. The cloud-based deployment ensures accessibility and scalability of the system. Overall, the project highlights how modern AI technologies can transform traditional construction management into a more automated, data-driven, and efficient process.

The project demonstrates an effective use of artificial intelligence in improving construction planning processes. The **Construction Planning Assistant Agent** reduces dependency on manual planning and enhances efficiency through automated task decomposition and intelligent decision-making. The integration of LLMs and multiple specialized agents ensures accurate and structured outputs for complex construction scenarios. The system is scalable, modular, and deployable in cloud environments, making it suitable for real-world applications. Overall, the project successfully achieves its objective of building an intelligent assistant that modernizes and optimizes construction project planning.

Chapter 8

Future Scope

- Integration of more advanced Large Language Models (LLMs) for improved accuracy and reasoning in construction planning.
- Development of a mobile application version for on-site project access and updates.
- Addition of real-time data integration from construction sites using IoT sensors.
- Implementation of advanced predictive analytics for better cost and risk forecasting.
- Expansion of the system to support large-scale enterprise project management with multi-user collaboration
- Addition of a chatbot and voice-enabled interface for more natural and accessible user interaction.

Appendix

The Appendix section provides additional supporting information related to the **Construction Planning Assistant Agent** project that helps in better understanding of the system implementation and usage.

Appendix A: Project Structure

```
construction-planning-assistant-agent/
│
├── app.py                    # Streamlit UI
├── main.py                   # Backend entry point
├── requirements.txt         # Dependencies
│
├── agents/
│   ├── planner_agent.py
│   ├── scheduling_agent.py
│   ├── resource_agent.py
│   ├── cost_agent.py
│   ├── risk_agent.py
│
├── utils/
│   ├── helper_functions.py
│
├── data/
│   ├── sample_inputs.json
│
└── Dockerfile               # Containerization file
```

Appendix B: Sample Input

- “Plan a residential building construction project for 6 months with limited budget and medium risk.”

Appendix C: Sample Output

- Project Schedule with phases (foundation, structure, finishing)
- Resource allocation plan (labor, materials, equipment)
- Estimated cost breakdown
- Risk analysis report with mitigation strategies

Appendix D: Tools Used

- Python 3.x
- Streamlit
- FastAPI / Flask
- Docker
- AWS EC2
- Git & GitHub
- Large Language Models (LLMs)

Appendix E: System Workflow Overview

User Input → UI (Streamlit) → Backend API → LLM Planner → Agents (Scheduling, Cost, Resource, Risk) → Aggregation → Final Output

Bibliography

The following references and resources were used for the development and understanding of the Construction Planning Assistant Agent project:

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